

Ternary liquid–liquid equilibrium of an azeotropic mixture (hexane + methanol) with different imidazolium-based ionic liquids at $T = 298.15$ K and 101.325 kPa

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ABSTRACT

The recovery and purification of methanol from its mixtures is of great importance; to this end, liquid–liquid extraction is an effective method for the separation of azeotropes. In this work, three imidazolium-based ionic liquids were used to separate a hexane–methanol azeotropic mixture, and the ternary liquid–liquid equilibria (LLE) data for the systems of hexane + methanol + [MIM][HSO₄] (1-methylimidazole hydrogen sulfate), [BMIM][HSO₄] (1-Butyl-3-methylimidazolium hydrogen sulfate), [BMIM][OTf] (1-Butyl-3-methylimidazolium trifluoromethanesulfonate) were measured at 298.15 K and atmospheric pressure. The separation factor and distribution coefficient were calculated from the experimental LLE data to evaluate the separation performance. The influence of the anions and mono-/di-substituted hydrocarbon chains on the imidazolium cation ring of the LLE was investigated. The experimental LLE data were successfully correlated by non-random two liquid (NRTL) and universal quasi-chemical (UNIQUAC) models, and the binary interaction parameters were obtained.

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1. Introduction

Liquid–liquid extraction is a widely used method for separating azeotropes and for certain extraction process, the determination of a suitable solvent is the most important factor. Ionic liquids (ILs) are of great interest and potential in the liquid–liquid extraction due to their excellent properties [1], such as low vapor pressure, non-flammability and chemical/thermal stability, which make them to a favorable solvent compared to traditional organic solvents [2,3]. Additionally, ILs are a “Designer Solvent,” which is a combination of an anion and cation; therefore, a variety of ILs can be synthesized for a particular purpose. Therefore, the primary objective is to know exactly what is being designed to meet the requirements of a specific separation process. In the last two decades, the applications of ILs as solvents have drawn extensive attention, in which many liquid–liquid equilibria systems containing ILs have been investigated [2–6].

There are a number of alcohol and alkane mixtures that can exist in the industrial production processes of oxygenated additives for unleaded gasoline [7,8]. Methanol is one of the basic organic raw materials in the chemical industry, in which it has been widely used in the organic synthesis industry, as well as acting as a good solvent. Therefore, the recovery of pure methanol from mixtures is of great economic significance. Due to the close boiling point and formation of azeotropes, ordinary distillation encounters difficulty in separating the mixtures. Liquid–liquid extraction is an effective method for the separation of azeotropes [9] and is an energy-saving process [10]. Liquid–liquid equilibrium (LLE) data of ternary systems are the primary consideration for the design of a separation process [11], in which a large number of LLE data are required for the separation of alcohol from its mixture.

Currently, research regarding the purification of alcohol by liquid–liquid extraction has included Gao et al., who studied the recovery of methanol/ethanol at different temperatures [12,13]. Marciniak et al. [14] analyzed the separation of methanol from alkanes using ILs as solvents, and Cai et al. [15] measured the LLE data for the ternary systems of methanol + hexane/heptane + IL.

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Table 1Sources, CAS number, molar mass, purification, water contents by mass, w_w , and analysis method of the chemicals used in this work.

Chemical	Source	CAS number	Molar mass/ (g·mol ⁻¹)	Mass purity stated by supplier	w_w / 10 ⁻⁶	Analysis method
ethanol	Tianjin Kermel Chemical Reagent Co., Ltd.	64-17-5	46.07	0.998 ^a		GC
hexane	Tianjin Kermel Chemical Reagent Co., Ltd.	110-54-3	86.18	0.995 ^a		GC
methanol	Tianjin Kermel Chemical Reagent Co., Ltd.	67-56-1	32.04	0.990 ^a		GC
[MIM][HSO ₄]	Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences.	681281-87-8	180.18	0.980 ^a	<500 ^a	
[BMIM][HSO ₄]	Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences.	262297-13-2	236.29	0.980 ^a	<500 ^a	
[BMIM][OTf]	Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences.	174899-66-2	288.29	0.980 ^a	<500 ^a	

^a Analysis by supplier.

Pereiro et al. [16,17] tested the recovery of ethanol from hexane/heptane by ILs and Revelli et al. [18], and Letcher et al. [19] reported the LLE for the mixtures of 1-methyl-3-octylimidazolium chloride + alkanol + alkane. Lastly, González et al. [20–22] published the LLE data for the ternary systems of alcohols + alkanes.

Imidazolium-based ILs are considered to have a moderate viscosity and low melting point [23,24]. In this work, three imidazolium-based ILs were studied for their use in the separation of methanol from a methanol-hexane mixture, and liquid-liquid equilibrium data for ternary systems of {hexane + methanol + [MIM][HSO₄], hexane + methanol + [BMIM][HSO₄], hexane + methanol + [BMIM][OTf]} were measured at 298.15 K (atmospheric pressure). Through the comparison of three ILs with different anions, we observed that the anion has a strong effect on the separation capacity. Two ILs with the same anionic substituents and mono-/di-substituted hydrocarbon chains on the imidazolium cation ring were applied in the separation process, and the influence of alkyl chains of the imidazole ring was discussed. Separation factors (S) and distribution coefficients (β) were calculated from the experimental tie-lines data to evaluate the separation capacity. The experimental tie-line data were correlated by use of the universal quasi-chemical (UNIQUAC) [25] and non-random two liquid (NRTL) [26] models, and the interaction parameters were obtained.

2. Experiment

2.1. Chemicals

Hexane, methanol and ethanol were provided by Tianjin Kermel Chemical Reagent Co., Ltd., with a purity of 99.5%, 99.5% and 99.8%, respectively, and all of the purities were checked by gas chromatography (GC-2014C). ILs were supplied by Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, with a stated purity of 98%. All chemicals were used without further purification. Detailed information, including source, purity and suppliers of chemical reagents, is shown in Table 1, and the ionic liquid structures were given in Fig. 1.

2.2. Apparatus and procedure

The LLE for the systems of (hexane + methanol + IL) was determined at 298.15 K and atmospheric pressure. The equipment and experimental procedure for the measurement of LLE data were detailed in our previous work [27–30]. The mixtures were prepared by mass and placed into the equilibrium cell [31], which was self-designed. The three components were mixed together and agitated by a stir bar. After intense agitation (3 h, approximately

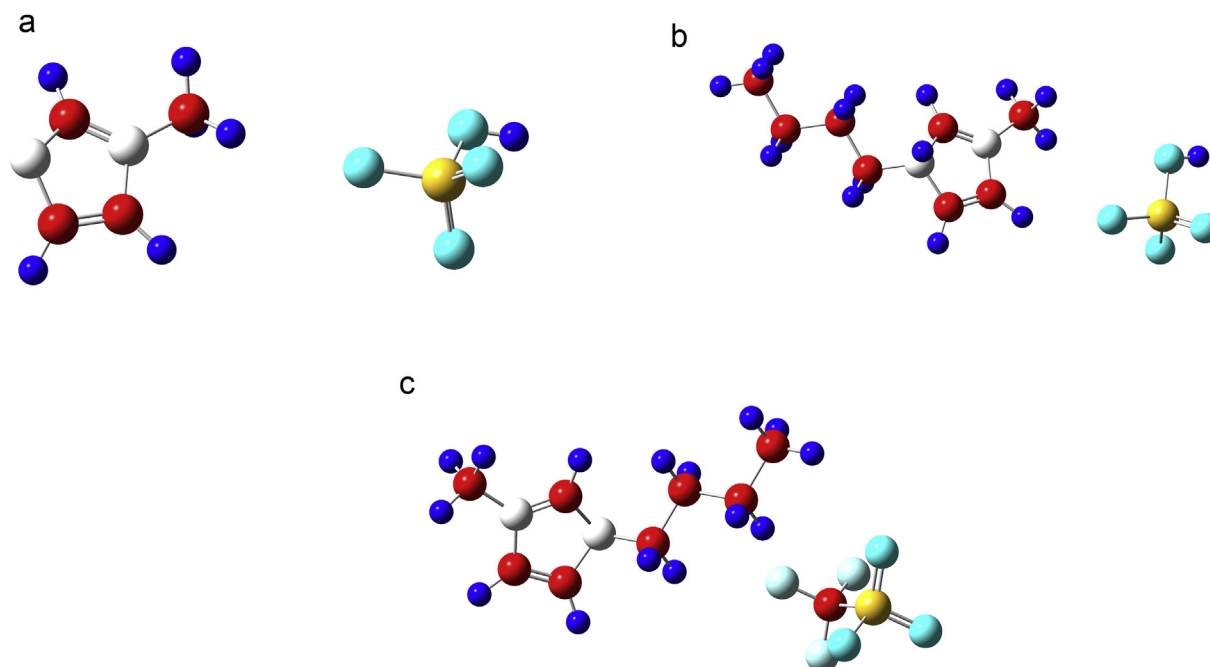
**Fig. 1.** Ionic liquids structures. (a) [MIM]HSO₄; (b) [BMIM]HSO₄; (c) [BMIM]OTf.

Table 2

Experimental LLE data on mole fraction x , distribution coefficient β and separation factor S for ternary systems of hexane (1) + methanol (2) + IL (3) at $T = 298.15$ K and $p = 0.1$ MPa.

Upper phase		Lower phase		β	S
x_1	x_2	x_1	x_2		
hexane (1) + methanol (2) + [MIM][HSO4] (3)					
0.9954	0.0037	0.0116	0.1171	31.6486	2715.7815
0.9872	0.0109	0.0160	0.3225	29.5872	1825.5275
0.9744	0.0231	0.0179	0.4532	19.6190	1067.9777
0.9650	0.0341	0.0168	0.5913	17.3402	996.0280
0.9517	0.0474	0.0156	0.6831	14.4114	879.1873
0.9391	0.0601	0.0137	0.7664	12.7521	874.1225
hexane (1) + methanol (2) + [BMIM][HSO4] (3)					
0.9963	0.0031	0.0082	0.1131	36.4839	4432.7903
0.9938	0.0053	0.0097	0.1906	35.9623	3684.4637
0.9871	0.0116	0.0103	0.3754	32.3621	3101.4173
0.9819	0.0164	0.0156	0.5360	32.6829	2057.1388
0.9739	0.0242	0.0200	0.7301	30.1694	1469.1000
0.9577	0.0400	0.0242	0.9239	23.0975	914.0692
hexane (1) + methanol (2) + [BMIM][OTf] (3)					
0.9962	0.0038	0.0290	0.1790	47.1053	1618.1470
0.9929	0.0070	0.0318	0.3075	43.9286	1371.5937
0.9878	0.0122	0.0326	0.5131	42.0574	1274.3643
0.9784	0.0216	0.0260	0.6409	29.6713	1116.5537
0.9731	0.0269	0.0263	0.7140	26.5428	982.0818
0.9681	0.0319	0.0329	0.7812	24.4890	720.6027
0.9531	0.0469	0.0309	0.8158	17.3945	536.5261

^a Standard uncertainties u are $u(x_i) = 0.003$, $u(T) = 0.05$ K, $u(p) = 0.0015$ MPa.

1000 r/min), the mixtures were left to settle (approximately 15 h) for complete separation. During the mixing and standing process, the temperature was maintained at 298.15 K using a low temperature thermostat (accurate to 0.05 K).

After separation of the two phases (hexane-rich phase and IL-rich phase), samples from each phase were taken to analyze the composition. Organic component composition was analyzed by GC using an internal standard method. The GC is equipped with a thermal conductivity detector (TCD) and a packed column of Porapak Q (connected after a pre-column); the carrier gas is helium (purity > 99.99%), and ethanol is the internal standard. The analyses were performed at least three times for each sample. The detecting condition was given in Table S1 (see the Supplementary Material). For ILs, a mass difference method was applied to analyze the composition. Weights of samples were measured twice before and after the evaporation of volatile components by a vacuum drying oven (DZF-6020, Shanghai Boxun, China), until the weights were constant; the temperature was set to 408.15 K.

3. Results and discussion

3.1. Experimental results

Measurements of LLE data for hexane (1) + methanol (2) + ILs (3) systems were carried out at 298.15 K and atmospheric pressure, the obtained results presented in Table 2; the tie-lines of the ternary systems were illustrated in Figs. 2–4. From the experimental phase diagrams, broad ranges of a liquid-liquid two-phase region can be observed for each system. Therefore, the separation of the hexane + methanol mixture using the selected ILs by liquid-liquid extraction can be feasible.

In the evaluation of the separation capacity of the ILs in separating the hexane + methanol mixture, the distribution coefficient (β) and separation factor (S) are two important parameters. The β refers to the solvent dosage, and S is relative to the product purity. A solvent with higher β and S values is more suitable for an extraction system. From the experimental data, the values of β and S were

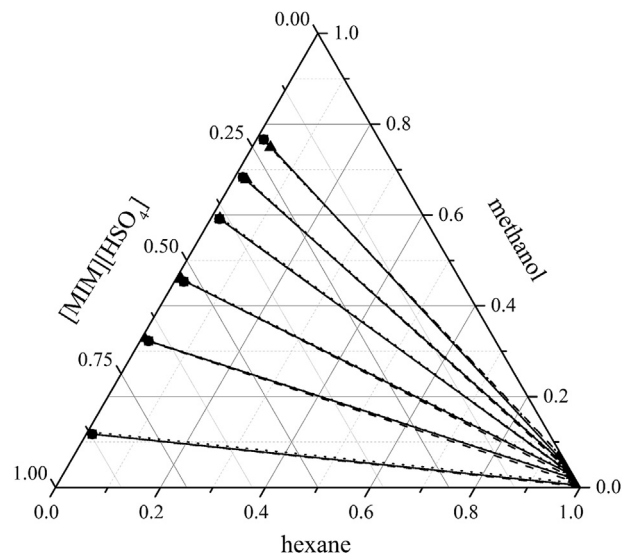


Fig. 2. Experimental and calculated LLE data in mole fraction for the system of hexane (1) + methanol (2) + [MIM][HSO₄] (3). (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model.

calculated for each system, and the equations are as follows:

$$\beta = \frac{x_2^{IL-rich}}{x_2^{hex-rich}} \quad (1)$$

$$S = \frac{(x_2^{IL-rich}/x_1^{IL-rich})}{(x_2^{hex-rich}/x_1^{hex-rich})} \quad (2)$$

where x presents the mole fraction, and subscripts 1 and 2 represent hexane and methanol, respectively. Superscript, hex-rich and IL-rich present hexane-rich phase and IL-rich phase, respectively. The calculated values of β and S for the (hexane + methanol + IL) systems were listed in Table 2 and illustrated in Figs. 5 and 6, respectively. The calculated values of β and S are considerably

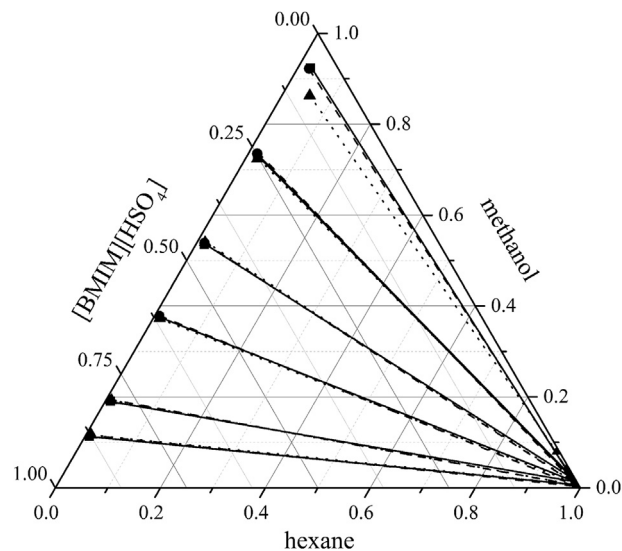


Fig. 3. Experimental and calculated LLE data in mole fraction for the system of hexane (1) + methanol (2) + [BMIM][HSO₄] (3). (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model.

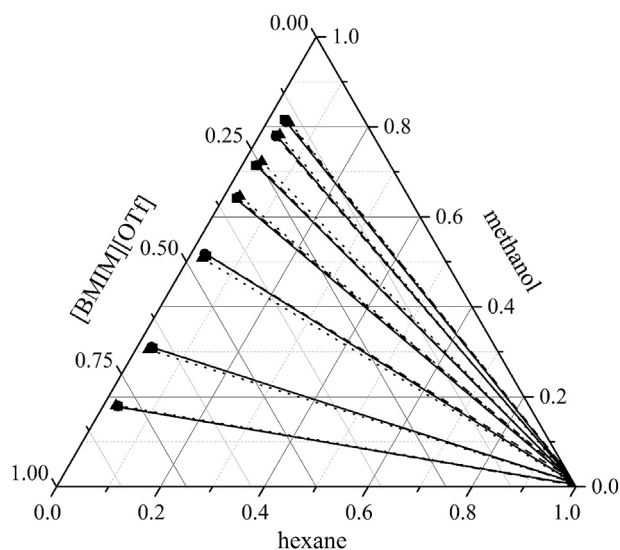


Fig. 4. Experimental and calculated LLE data in mole fraction for the system of hexane (1) + methanol (2) + [BMIM] [OTf] (3). (■), experimental value; (●), calculated value by NRTL model; (▲), calculated value by UNIQUAC model.

higher than the unity, indicating that all selected ILs are effective solvents for the extraction of methanol from hexane. Therefore, it can be concluded that methanol has a greater affinity for the selected ILs compared with hexane. Additionally, the values of β and S showed a decreasing trend with the increase of $x_{IL-rich}^2$. For ILs that have the same cation, [BMIM][HSO₄] has higher S values than [BMIM][OTf] in the overall concentration range. Therefore, the extraction capacity for the anion follows the trend of [HSO₄][−] > [OTf][−]; the anion, [HSO₄][−], has a greater affinity for methanol. [BMIM][HSO₄] has larger S values than [MIM][HSO₄]; thus, di-substituted ILs (di-substituted hydrocarbon chains on the imidazolium cation ring) are more suitable for the separation of the hexane + methanol mixture, compared with mono-substituted ILs (mono-substituted hydrocarbon chains on the imidazolium cation ring).

Cai et al. has measured the LLE data for the ternary systems of {hexane + methanol + [MMIM][DMP]}/[EMIM][DEP]/[BMIM][DBP] and the systems of {hexane + methanol + [BTMA][NTf₂]} / [BTMA]

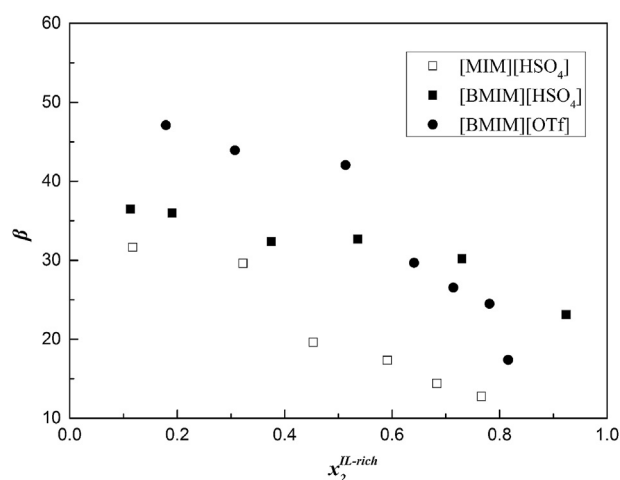


Fig. 5. Distribution coefficient (β) for ternary systems of for the systems of hexane + methanol + IL, $x_{IL-rich}^2$ is the mole fraction of methanol in IL-rich phase. (■), hexane + methanol + [MIM] [HSO₄]; (□), hexane + methanol + [BMIM] [HSO₄]; (●), hexane + methanol + [BMIM] [OTf].

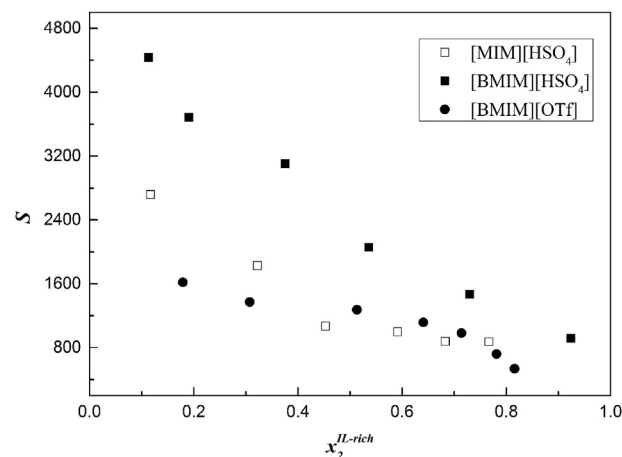


Fig. 6. Separation factor (S) for ternary systems of for the systems of hexane + methanol + IL, $x_{IL-rich}^2$ is the mole fraction of methanol in IL-rich phase. (■), hexane + methanol + [MIM] [HSO₄]; (□), hexane + methanol + [BMIM] [HSO₄]; (●), hexane + methanol + [BMIM] [OTf].

[NTf₂]} measured by Aranda et al.; comparisons with our data were given in Fig. S1 and Fig. S2 (see Supplementary Material). The values of β and S are larger than all ILs combined. As shown in Fig. S1, for all systems of {hexane + methanol + ILs}, [MMIM][DMP] showed the highest values of β , which means that a smaller amount of solvent is required for the separation of the hexane + methanol mixture. In Fig. S2, [BMIM][HSO₄] showed the highest values of S ; therefore, [BMIM][HSO₄] can be a good choice for high purity methanol.

3.2. Sigma profiles

The conductor-like screening model COSMO-SAC [32–34] was used to calculate the sigma profiles, which is a descriptor for the local polarity of molecular surface and determines the interaction energies. The sigma profiles for the studied components are shown in Fig. 7. Two important vertical dashed lines, $\sigma = -0.0082 \text{ e}/\text{\AA}^2$ and $\sigma = +0.0082 \text{ e}/\text{\AA}^2$, divided the sigma profile into three areas. Between the two vertical dashed lines is a neutral zone. The left side of $\sigma = -0.0082 \text{ e}/\text{\AA}^2$ is the hydrogen donor zone and the right side of $\sigma = +0.0082 \text{ e}/\text{\AA}^2$ is hydrogen an acceptor zone. It seen from Fig. 7, hexane is almost completely in the middle of the area, and

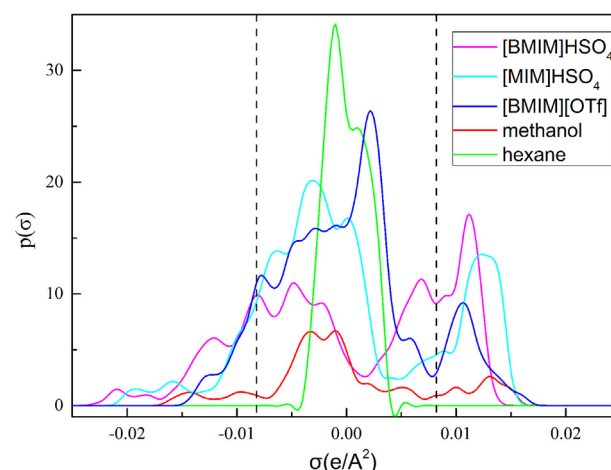


Fig. 7. Sigma profile for [MIM]HSO₄, [BMIM]HSO₄, [BMIM][OTf], hexane and methanol.

the methanol compound is a small fraction of the profile, which lies in the hydrogen donor zone and hydrogen acceptor zone. Therefore, the extraction ability of ILs will depend on the hydrogen bond donor or acceptor ability; [MIM]HSO₄ and [BMIM][OTf] shows a similar distribution fraction. The profile of [BMIM]HSO₄ has larger fraction, which lies in the hydrogen donor zone and hydrogen acceptor zone, than does [MIM]HSO₄ and [BMIM][OTf]. Therefore, it was speculated that [BMIM]HSO₄ is the most suitable solvent to extract methanol from hexane, which is consistent with the experimental results.

3.3. LLE correlation

Two models were adopted to correlate the experimental data and regress the binary interaction parameters. The NRTL model (entropic influence) is the native composition theory based on activity coefficient, which is led by the shape properties and molecule. The UNIQUAC model (enthalpic influence) accounts for the energy relations, which contains of a combinatorial and remaining involvement for the excess Gibbs energy. The r and q values for the UNIQUAC model were obtained from the literature [35,36]; the values were listed in Table 3. The molecular interactions were studied for the systems of {hexane + methanol + [MIM][HSO₄], hexane + methanol + [BMIM][HSO₄] and hexane + methanol + [BMIM][OTf]} from the LLE values. The binary interaction parameters Δg_{ij} and non-randomness factor α for the NRTL model, and the binary interaction parameters Δu_{ij} for UNIQUAC model are regressed from the experimental LLE data by minimizing the mole fraction differences between the experimental and calculated, and the results were given in Table 4. The root-mean-square deviations ($rmsd$) between the experimental and calculated values were determined and given in Table 4; the objective function (OF) and $rmsd$ equations are expressed as follows:

$$OF = \sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 (x_{ijk}^{exp} - x_{ijk}^{calc})^2 \quad (3)$$

$$rmsd = 100 \left(\sum_{k=1}^M \sum_{j=1}^2 \sum_{i=1}^3 \frac{(x_{ijk}^{exp} - x_{ijk}^{calc})^2}{6M} \right)^{1/2} \quad (4)$$

where M indicates the number of tie-lines, x_{ijk}^{exp} and x_{ijk}^{calc} represent the experimental and calculated mole fraction, respectively, and subscripts i , j , and k indicate component, phase, and tie-line, respectively. The tie-line data, calculated by NRTL and UNIQUAC models, are shown in Figs. 2–4, accordingly. Reasonable correlations were obtained, in which $rmsd$ are smaller than 0.016 for all systems, which can be observed in Figs. 2–4.

Table 3
Volume (r_i) and surface area (q_i , q_i') of UNIQUAC equation.

Component	r_i	q_i	q_i'
methanol ^a	1.4311	1.432	0.96
hexane ^b	4.500	3.856	3.856
[MIM][HSO ₄] ^b	5.529	4.505	4.505
[BMIM][HSO ₄] ^b	8.2203	6.628	6.628
[BMIM][OTf] ^a	12.460	7.518	7.518

^a From reference [35].

^b From reference [36].

Table 4

Binary interaction parameters Δg_{ij} , Δg_{ji} and nonrandomness factor α for NRTL model, binary interaction parameters Δu_{ij} and Δu_{ji} for UNIQUAC model and root-mean-square deviations, obtained from the experimental LLE data of studied ternary systems by NRTL and UNIQUAC models at $T = 298.15$ K^a, $p = .1$ MPa.

i - j	Δg_{ij} (kJ·mol ⁻¹)	Δg_{ji} (kJ·mol ⁻¹)	$rmsd$	α
NRTL parameters				
hexane (1) + methanol (2) + [MIM][HSO ₄] (3)			0.0071	0.20
1-2	3.1503	3.4498		
1-3	11.7138	13.7594		
2-3	11.1783	-10.2185		
hexane (1) + methanol (2) + [BMIM][HSO ₄] (3)			0.0081	0.20
1-2	1.2862	1.3735		
1-3	3.9104	5.9052		
2-3	1.3692	10.0138		
hexane (1) + methanol (2) + [BMIM][OTf] (3)			0.0044	0.20
1-2	3.1849	3.4093		
1-3	15.3083	4.7397		
2-3	17.0350	-15.4426		
i - j	Δu_{ij} (kJ·mol ⁻¹)	Δu_{ji} (kJ·mol ⁻¹)	$rmsd$	
UNIQUAC parameters				
hexane (1) + methanol (2) + [MIM][HSO ₄] (3)			0.0045	
1-2	12.5334	0.0447		
1-3	2.3377	1.7636		
2-3	0.0560	1.8422		
hexane (1) + methanol (2) + [BMIM][HSO ₄] (3)			0.0160	
1-2	13.0559	0.0450		
1-3	2.5545	1.4269		
2-3	0.0583	0.9451		
hexane (1) + methanol (2) + [BMIM][OTf] (3)			0.0056	
1-2	12.3945	0.0483		
1-3	5.9202	0.2065		
2-3	0.0494	0.5990		

^a Standard uncertainties u are $u(T) = 0.05$ K.

4. Conclusions

Experimental LLE data for the ternary systems of methanol + hexane + imidazolium-based ILs were determined at 298.15 K and atmospheric pressure, with the systems of hexane + methanol + [MIM][HSO₄], [BMIM][HSO₄] and [BMIM][OTf] exhibiting good separation performance. The values of β and S were calculated from the experimental tie-lines data. In the studied concentration range, the values β and S are considerable high, especially in low methanol concentrations. Therefore, the three studied ILs are a suitable solvent in the purification of methanol. In comparison of the separation performance for the three ILs in the separation of the hexane-methanol mixture, it can be concluded that anion had a significant effect on the separation performance; the anion [HSO₄][–] had the maximum extraction potential. Di-substituted ILs are more suitable for the hexane-methanol separation compared with mono-substituted ILs. The NRTL and UNIQUAC thermodynamics models were adopted to correct the experimental tie-lines, and satisfactory results were obtained for all of the studied systems.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.fluid.2018.01.008>.

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